Bargaining over Cabinets: Multi-Agent RL for Portfolio Allocation in Coalition Governments

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GitHub: `https://github.com/bot2048/cabinet-bargaining-marl`

1 Problem

After an election without a majority, parties bargain over who governs and how ministries are divided. Real cabinets split portfolios almost exactly in proportion to seats (*Gamson's law*), yet the standard bargaining theory (Baron–Ferejohn, BF) predicts that the *formateur*, the party that proposes the cabinet, captures a premium. We model parties as learning agents in an environment that encodes the rules of government formation, *validate* the learners against the closed-form BF equilibrium, and then switch institutional mechanisms on one at a time to see which of them push learned allocations towards proportionality. Agents learn only from environment rewards; no fairness term is built in, so any proportionality must emerge.

2 RL formulation

Environment (`cabinet/env.py`). A partially observable stochastic game with n parties (seats w_i, quota $q = \lfloor W/2 \rfloor + 1$) and m ministries. Each period without a cabinet is a formation round: nature draws a formateur $f \sim \rho$; f proposes a winning coalition $C \ni f$ and an allocation $\mu : \{1..m\} \to C$; all partners vote simultaneously and the cabinet forms only if all accept. A sitting cabinet pays per-period flows; members may walk out (cabinet collapses). Optional mechanisms: repeated terms with elections (M1), private types (M4), heterogeneous ministry values (M5), exits (M8), ministerial policy (M9), seat-proportional recognition (M10). Levels E0–E6 are parameter restrictions of one class (`cabinet/config.py`); E0 is exactly the symmetric BF game.

State / observation space $\mathcal{S}$. Agent i sees a *set* of n party tokens (26 features: seat share, recognition probability, self/formateur/member flags, pending and current shares, history of the last Λ cabinets), m ministry tokens (8: salience, own valuation, policy dimension led), $m \times n$ pair features (4: partners' valuations and policy costs when types are public), a context vector (17: phase, quota, δ, horizon, λ, ...) and 8 decision-specific features (offer, exit bonus). Tokens carry no party labels, so one network handles any $n \in \{3..12\}$, $m \le 25$ and is permutation-equivariant.

Action space $\mathcal{A}$ (masked, mixed discrete). Formateur: $n-1$ binary include/exclude decisions (masks guarantee the coalition stays winning; $|\mathcal{A}|$ up to 10^{19} proposals is decoded in $n-1+m$ small steps), then for each ministry a pointer over coalition members, or, for a divisible surplus, an offer on a 1% grid (101 bins) per partner. Voter: accept/reject. Member: stay/exit.

Reward $\mathcal{R}$. In a governed period, member i receives

$$r_i = \textstyle\sum_{k:\mu(k)=i} \nu_{ik} \;-\; \lambda \frac{1}{d} \sum_{h=1}^{d} |y_h - z_{ih}|,$$

office value of held ministries minus a policy loss (minister of the lead ministry sets policy y_h); outsiders get only the policy term, an exiting member its bonus e_i; failed rounds pay nothing. Discount δ per period. With $\lambda = 0$ the game is constant-sum up to delay, so a team reward would erase the conflict under study: every agent maximises its *own* return.

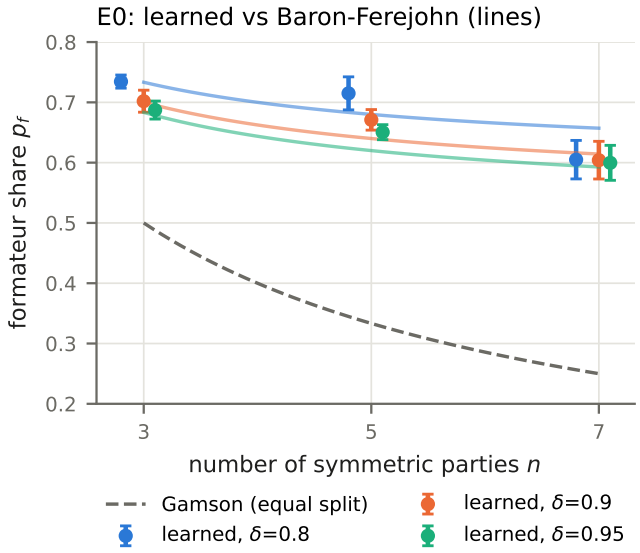


Figure 1: Benchmark recovery (E0). Lines: closed-form BF formateur share; dots: learned (mean over 6 seeds). Dashed: Gamson's equal split.

Algorithm (`cabinet/policy.py`, `cabinet/ppo.py`). One shared network (set encoder: 2 attention blocks over parties with cross-attention to ministries) with masked autoregressive proposal heads, vote/exit heads and a value head. Each agent acts only at some steps, so we run PPO with GAE on every agent's own decision sequence (semi-MDP: interval rewards and discounts between its decisions). Regimes: LA1 shared-parameter IPPO self-play; LA2 MAPPO with a centralised critic that sees all private types; Fict. self-play (LA3) against a pool of past checkpoints. Rollouts run on CPU, updates on GPU. **Baselines** (`cabinet/baselines.py`): BL1 exact BF equilibrium strategy (from our certified solver, `cabinet/bf_solver.py`), BL2 Gamson agent (minimal-seat coalition, proportional split), BL3 myopic BF heuristic, BL4 random feasible agent.

Metrics (`cabinet/metrics.py`). Formateur premium $\Phi = \mathbb{E}[p_f - s_f]$ and Gamson distance $D_G = \mathbb{E}[\frac{1}{2}\sum_i |p_i - s_i|]$ (s_i: seat share within the coalition), both *floor-adjusted* ($\tilde{\Phi}, \tilde{D}_G$) against the most proportional feasible assignment (Hamilton's rule), the BF gap $\Delta_{\mathrm{BF}} = \mathbb{E}[p_f] - x_f^{\mathrm{BF}}$, the regression $p = \alpha + \beta s + \gamma F$, and exploitability (gain of a trained best response, per period).

3 Experiments and results

Setup. Scaled-down sizes for one laptop (i9 CPU + RTX 4060): $n \le 5$ parties (3–8 for real parliaments), $m = 6$ ministries, 6-period terms, 3 expected terms, 500–700 PPO iterations (≈4–6M decisions) per run, ≥ 3 seeds per cell (10 where the success criteria demand it). Every configuration is a parameter restriction of one environment; mechanisms are switched on one at a time from a base game (BF with indivisible ministries) and off one at a time from E* (all on).

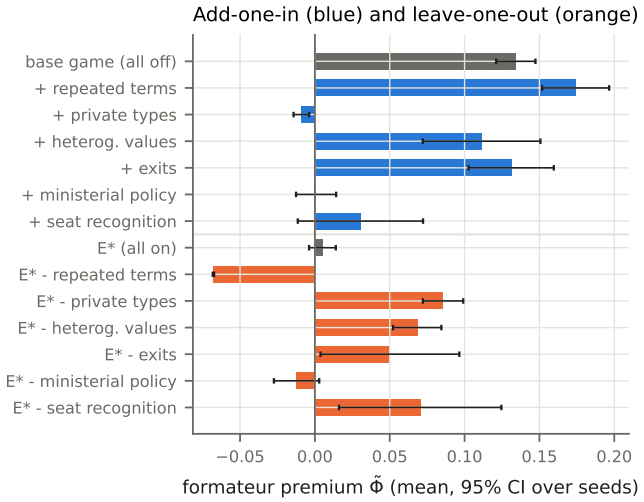


Figure 2: Floor-adjusted formateur premium $\tilde{\Phi}$ and Gamson distance $\tilde{D}_G$ per configuration (mean, 95% CI over seeds).

Table 1: E* under different learners and memory (mean over seeds).

	$\tilde{\Phi}$	$\tilde{D}_G$	$\hat{\beta}$	$\hat{\gamma}$	Φ_ν
IPPO, window Λ=4	+0.005	+0.097	+0.401	+0.079	+0.155
MAPPO (LA2)	-0.030	+0.089	+0.458	+0.007	+0.124
Fict. self-play (LA3)	-0.002	+0.097	+0.402	+0.063	+0.150
IPPO + GRU memory	–	–	–	–	–
IPPO, memoryless	-0.043	+0.098	+0.430	-0.008	+0.111

Validation against theory (RQ1). In the symmetric BF game the learned formateur share is within 0.019 of the closed form on average (mean gap +0.005; 5/9 cells within the 0.02 tolerance; failure rate 1.3%), Fig. 1. Errors grow with n: with more partners each voter's threshold is learned less precisely. With weighted seats (E1) the gap to the certified solver is -0.073. The pipeline therefore reproduces BF bargaining, the precondition for trusting it elsewhere.

Which mechanisms remove the formateur premium (RQ2). The base game keeps a clear premium ($\tilde{\Phi} = +0.134$, regression $\hat{\gamma} = +0.231$). Switched on alone (Fig. 2, blue), *private types* ($\tilde{\Phi} = -0.009$) and *ministerial policy* (+0.001) are each sufficient to remove it; seat-proportional recognition nearly is (+0.030); repeated terms (+0.174), exits (+0.131) and heterogeneous values (+0.111) are not. Repetition fails because agents discount the next term heavily; with patient agents (cross-term discount 0.78 instead of 0.35) it gives – (– with GRU memory). Switched off one at a time from E* (orange), removing private types brings the premium back (+0.086), so private information is also *necessary*; removing heterogeneous values, exits or seat recognition partly restores it. In E* the premium is gone ($\tilde{\Phi} = +0.005$, $\hat{\gamma} = +0.079$) but *moves into quality*: in the formateur's own valuation it is +0.155, i.e. the formateur takes the ministries it values most rather than more of them, as the empirical literature has hypothesised. Allocations are flatter than seats ($\hat{\beta} = +0.401$), the small-party bias of real cabinets, and voting power explains shares better than seats (R^2 0.56 vs 0.43 for Banzhaf vs seat shares; H4). The Gamson region ($\tilde{D}_G \leq 0.05$) is not reached ($\tilde{D}_G = +0.097$).

Learners, memory and equilibrium checks (RQ5–6). Table 1: the result holds across learning regimes and memory variants, and symmetric PSRO (LA4) reaches $\tilde{\Phi} = --$ on E*. Cross-play mismatch is – within and – across algorithms per period, so the runs reach compatible conventions. Trained best responses gain +0.012 (E*, IPPO), – (E*, MAPPO) and +0.017 (base) office units per period, the order of the 0.01 tolerance; but they also barely exploit a scripted Gamson profile (+0.008), so at our budget the test has little power and we claim approximate, not certified, equilibrium play. A higher entropy temperature (bounded rationality, H9) lowers the base premium only slightly (+0.141 to +0.125).

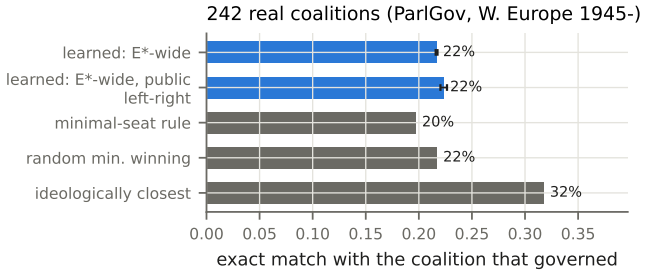


Figure 3: Real parliaments: coalition proposed by the real prime minister's party vs the coalition that governed.

Real-world test (RQ4, RQ7). Policies trained only on random parliaments were placed into the 242 post-1945 Western European majority coalitions of ParlGov, with the real prime minister's party as formateur (Fig. 3). The learned formateur proposes the coalition that actually governed in 21.7% of cases, versus 19.7% for Riker's minimal-seat rule, 21.7% for a random minimal winning coalition and 31.7% for the ideologically closest one, which uses left–right scores the agents cannot see; with public positions set from those scores it reaches 22.3%. Across 441 coalitions in 33 parliamentary democracies: 19.7% (minimal-seat 20.4%, ideology 31.9%). Agents thus match size-based theories but miss ideological connectedness. On real seat distributions the learned allocations again show no premium ($\tilde{\Phi} = -0.034$, $\hat{\gamma} = +0.050$) and follow voting power more than seats (R^2 0.73 vs 0.64).

4 Conclusions and limitations

Learning parties reproduce Baron–Ferejohn bargaining where theory applies, and two features absent from that model, private information and policy motivation, are each enough to make the formateur premium vanish, as in real cabinets, with private information also necessary; the premium survives in the *quality* of portfolios. Limitations: scaled-down sizes and budgets; exploitability tests can refute but not prove equilibrium play; the stored-state GRU is trained without back-propagation through time; opponent shaping (LA5) and the demand/pre-electoral modules were not run; ParlGov has no portfolio counts, so real allocations are compared only through coalitions. Coverage of every specification item: `docs/SPEC_COVERAGE.md`.

Key code. Environment `cabinet/env.py`; levels and mechanisms `cabinet/config.py`; policy `cabinet/policy.py`; PPO, MAPPO, FSP, exploitability `cabinet/ppo.py`; BF solver `cabinet/bf_solver.py`; metrics `cabinet/metrics.py`; baselines `cabinet/baselines.py`; experiments `scripts/run_experiment.py`; real parliaments `scripts/parlgov_eval.py`; tests `tests/test_core.py`.

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